Yixing Wu

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Publications

A Green Learning Approach to LDCT Image Restoration
IEEE International Conference on Image Processing (ICIP), 2025, pp. 1762–1767

A Statistical Approach for Modeling Irregular Multivariate Time Series with Missing Observations
APSIPA (Under Review)

Prediction of PD-L1 and CD68 in Clear Cell Renal Cell Carcinoma with Green Learning
Journal of Imaging 11 (6), 191

A Statistics-based Feature Generation (SFG) Method: Theory and Applications
2024 IEEE International Conference on Big Data (BigData), 5731-5738

Discriminant Radiomic Feature Selection for PD-L1 Prediction in Clear Cell Renal Cell Carcinoma
2023 19th International Symposium on Medical Information Processing and Analysis

EDUCATION

University of Southern California — **Sep 2024 – Now**
Ph.D. in Electrical Engineering — *Los Angeles, CA*

University of Southern California — **Jan 2022 – Dec 2023**
M.S. in Electrical Engineering, GPA 3.95/4.0 — *Los Angeles, CA*

Xidian University — **Sep 2016 – Jul 2020**
B.S. in Information Engineering, GPA 3.5/4.0 — *Xi'an, China*

WORK EXPERIENCE

USC Media Communications Lab — **Jun 2023 – Present**
Research Assistant — *Los Angeles, CA*

- Utilized Green Learning methods to improve the prediction performance of PD-L1 levels in clear cell renal cell carcinoma (ccRCC), enhancing non-invasive tumor prognosis via CT radiomics.
- Further extended model capabilities to address multiclass classification challenges for PD-L1 levels in ccRCC, demonstrating robust classification proficiency across imbalanced datasets.

Chinese Academy of Science — **Jun 2021 – Nov 2021**
Research Assistant — *Beijing, China*

- Designed the FPGA modules of frame analysis, frame assembly, synchronization between grandmaster and slave devices based on 802.1AS protocol using C++

PROJECTS

Course Project from EE 569 — **Jun 2023**

- Executed image demosaicing techniques and enhanced visual output through histogram manipulation.
- Employed the Sobel Edge Detector for precise edge detection within images.
- Integrated texture analysis methodologies using Law's filter and K-means clustering for advanced image segmentation.

Labs from CS231n: Convolutional Neural Networks for Visual Recognition — **March 2023**

- Implemented and applied kNN, SVM, Softmax classifiers.
- Implemented and applied vectorized backpropagation, Batch Normalization and Layer Normalization and Dropout.
- Implemented and applied RNN and Transformer networks, and Generative Adversarial Network (GAN).

Solar Panel Security System — **October 2022**

- Designed and built a solar panel motion detection system by using GPS, IMU sensors and xDot
- Analyzed data through AWS IOT Core, AWS Lambda and AWS dynamoDB
- Deployed a website on AWS EC2 using **Python Flask**, and **Google Map API**

A Cloud Storage & File Sharing Platform | github.com/star-wyx/drive **December 2021**

- Designed and built a **complete** web cloud storage system
- Implemented the features of Online preview, MD5 Back-End file storage system, Piazza for file sharing and Chat room.
- Implemented **APIs** using **Springboot**, and utilized **MongoDB** as database.
- Deployed the service to **Docker**

A Reliable File Transfer Protocol | github.com/star-wyx/file-protocol **Sept 2021**

- Collaborated with 2 teammates to design a reliable file transfer protocol based on **C++**
- Implemented with UDP socket to send packets and TCP socket to send ACK
- 10 times faster than scp protocol at high latency and low bandwidth network
- Deployed the service to **AWS EC2**